

Week 11

Thursday, 27 June 2024

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Tutorial: SQL – Basic SELECT statement

- 1) What are the two major components of SQL and what function do they serve?
- 2) Explain the function of each of the clauses in the basic SELECT statement.
 - Select - specifies which columns are to appear in the output
 - From - specifies the table to be used
 - Inner Join..ON - specifies additional table to be used. Repeat for each additional table
 - Where - filters the rows subject to some condition
 - Order by - specifies the order of the rows in the output
- 3) Refer to the HR Schema in the Appendix, Write SQL query that fulfill the requirements:
 - a) Display employee's first name, last name and department ID.
 - b) Display the employee's full name (first name join with last name) and department ID.

EMPLOYEE_ID	LAST_NAME	FIRST_NAME	DEPARTMENT_ID
1	Donald	OConnell	50
2	Douglas	Grant	50
3	Jennifer	Whalen	10
4	Michael	Hartstein	20

Select concat(first_name, ' ', last_name) as Full_name, department_id
From employees;

onal tab;e

c_name)
d

- c) For employees in Department 20, list the employees in the following format:
<employee's full name> was hired on <hire date>.

	EMP_HIRE_DATE
1	Michael Hartstein was hired on 17/02/04
2	Pat Fay was hired on 17/08/05

Select concat(first_name, ' ', last_name) as emp_full_name,
emp_hire_date
From employees
Where department_ID = 20;

- d) For employees whose phone number starts with 515, Display their first name, department ID and the following phone numbers. Name the new column as phone num no area code. Sort the result according to department id.

	FIRST_NAME	DEPARTMENT_ID	PHONE_NUM_NO_AREA_CODE
1	Jennifer	10	123.4444
2	Michael	20	123.5555
3	Guy	30	127.4565
4	Karen	30	127.4566
5	Sigal	30	127.4564
6	Shelli	30	127.4563

Select first_name, department_id,
substr(phone_number, 10) as phone_num_no_area_code
From employees
Where phone_number like '515%';
Order by department_id;

- e) For employees that have commissions, calculate the annual gross salary (salary + salary * commission_pct) * 12). List the employee's first name, department id, and the annual gross salary according to ascending order of the department id, then by descending order of the annual salary.

	FIRST_NAME	DEPARTMENT_ID	ANNUAL_GROSS_SALARY
1	John	80	235200
2	Karen	80	210600
3	Alberto	80	187200
4	Lisa	80	172500

Salary
(Salary * 12)*(1+pct)

- f) List the employees who join the company for more than 10 years. List the full name and the hire date. Most senior staff come first.

	FULL_NAME	HIRE_DATE
1	Lex, De Haan	13/01/01

Select concat(first_name, ' ', last_name) as full_name,
hire_date
From employees
Where (sysdate - hire_date) / 365 > 10;
Order by full_name;

st_name, ' ' was hired on', hire_date)

partment_id,
er,5,8) phone_num_no_area

er like '515%'

t_id :

```
Select first_name, department_id  
((salary*(1+commission_pct))*12 as  
annual_gross_salary  
From employees  
Where comission_pct is not null  
Order by department_id,  
annual_gross_salary
```

4	Hermann, Baer	07/06/02
3	Shelley, Higgins	07/06/02
4	William, Gietz	07/06/02
5	Susan, Mavris	07/06/02
6	Daniel, Faviet	16/08/02
7	Nancy, Greenberg	17/08/02
8	Den, Raphaely	07/12/02
9	Payam, Kaufling	01/05/03

full_name, hire_date

From employees

Where year(curdate()) - year(hire_date) > 10

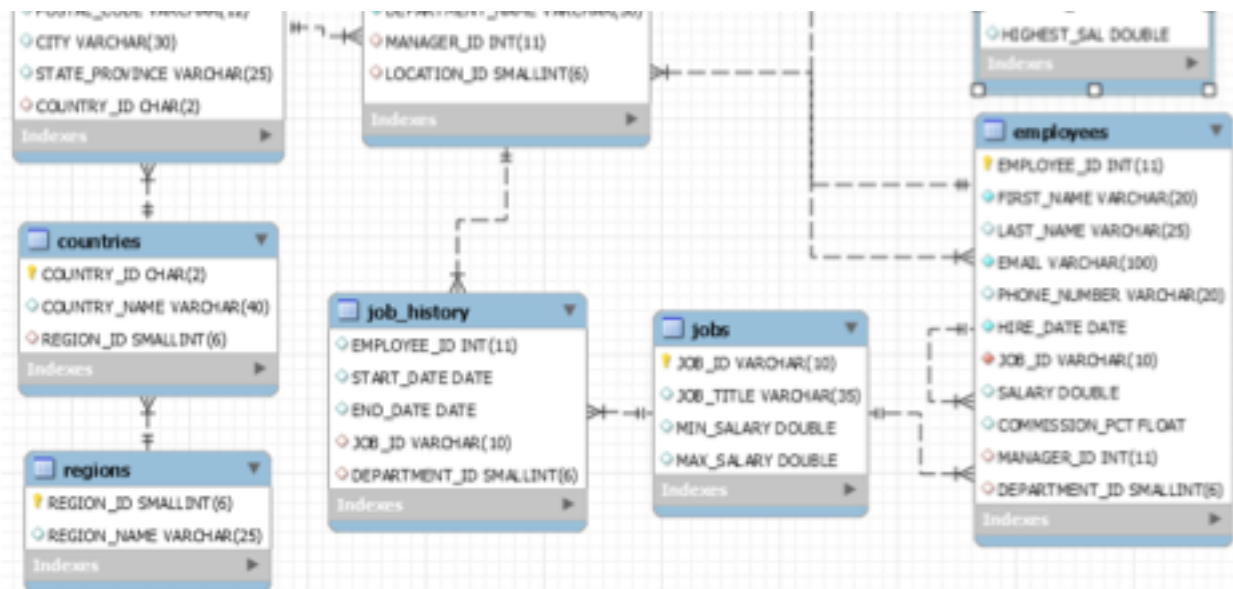
Orderby hire_date;

- g) List without repetition the country code in the company's employee list.
- h) List the employees' first name, last name, and department from following countries: CA, DE, and UK.
- i) List the employee's full name and their manager's full name for department 20. (self-inner-join)

Appendix - HR database relational schema

Schema





Sample Data:

COUNTRIES

COUNTRY_ID	COUNTRY_NAME	REGION_ID
1 AR	Argentina	2
2 AU	Australia	3
3 BE	Belgium	1
4 BR	Brazil	2
5 CA	Canada	2
6 CH	Switzerland	1
7 CN	China	3
8 DE	Germany	1
9 DK	Denmark	1
10 EG	Egypt	4
11 FR	France	1
12 IL	Israel	4
13 IN	India	3
14 IT	Italy	1
15 JP	Japan	3
16 KW	Kuwait	4
17 ML	Malaysia	3
18 MX	Mexico	2
19 NG	Nigeria	4
20 NL	Netherlands	1
21 SG	Singapore	3
22 UK	United Kingdom	1
23 US	United States of America	2
24 ZM	Zambia	4
25 ZW	Zimbabwe	4

DEPARTMENTS

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	Administration	200	1700
2	20	Marketing	201	1800
3	30	Purchasing	114	1700
4	40	Human Resources	203	2400
5	50	Shipping	121	1500
6	60	IT	103	1400
7	70	Public Relations	204	2700
8	80	Sales	145	2500
9	90	Executive	100	1700
10	100	Finance	108	1700
11	110	Accounting	205	1700
12	120	Treasury	(null)	1700
13	130	Corporate Tax	(null)	1700
14	140	Control And Credit	(null)	1700
15	150	Shareholder Services	(null)	1700
16	160	Benefits	(null)	1700
17	170	Manufacturing	(null)	1700
18	180	Construction	(null)	1700
19	190	Contracting	(null)	1700
20	200	Operations	(null)	1700
21	210	IT Support	(null)	1700
22	220	NOC	(null)	1700
23	230	IT Helpdesk	(null)	1700
24	240	Government Sales	(null)	1700
25	250	Retail Sales	(null)	1700
26	260	Recruiting	(null)	1700
27	270	Payroll	(null)	1700

EMPLOYEES

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	198 Donald	OConnell	DOCONNEL	650.507.9833	21/06/07	SH_CLERK	2600	(null)	124	50
2	199 Douglas	Grant	DGRANT	650.507.9844	13/01/08	SH_CLERK	2600	(null)	124	50
3	200 Jennifer	Whalen	JWHALEN	515.123.4444	17/09/03	AD_ASST	4400	(null)	101	10
4	201 Michael	Hartstein	MHARTSTE	515.123.5555	17/02/04	MK_MAN	13000	(null)	100	20
5	202 Pat	Fay	PFAY	603.123.6666	17/08/05	MK_REP	6000	(null)	201	20
6	203 Susan	Mavris	SMAVRIS	515.123.7777	07/06/02	HR_REP	6500	(null)	101	40
7	204 Hermann	Beer	HBAER	515.123.8888	07/06/02	PR_REP	10000	(null)	101	70
8	205 Shelley	Higgins	SHIGGINS	515.123.8080	07/06/02	AC_MGR	12000	(null)	101	110
9	206 William	Gietz	WGIEZT	515.123.8181	07/06/02	AC_ACCOUNT	8300	(null)	205	110
10	100 Steven	King	SKING	515.123.4567	17/06/03	AD PRES	24000	(null)	(null)	90
11	101 Neena	Kochhar	NKOCHHAR	515.123.4568	21/09/05	AD_VP	17000	(null)	100	90
12	102 Lex	De Haan	LDEHAAN	515.123.4569	13/01/01	AD_VP	17000	(null)	100	90
13	103 Alexander	Hunold	AHUNOLD	590.423.4567	03/01/06	IT_PROG	9000	(null)	102	60
14	104 Bruce	Ernst	BERNST	590.423.4568	21/05/07	IT_PROG	6000	(null)	103	60
15	105 David	Austin	DAUSTIN	590.423.4569	25/06/05	IT_PROG	4800	(null)	103	60
16	106 Valli	Pataballa	VPATABAL	590.423.4560	05/02/06	IT_PROG	4800	(null)	103	60
17	107 Diana	Lorentz	DLORENTZ	590.423.5567	07/02/07	IT_PROG	4200	(null)	103	60

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EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
55	145 John	Russell	JRUSSEL	011.44.134...	01/10/04	SA_MAN	14000	0.4	100	80
56	146 Karen	Partners	KPARTNER	011.44.134...	05/01/05	SA_MAN	13500	0.3	100	80
57	147 Alberto	Errazuriz	AERRAZUR	011.44.134...	10/03/05	SA_MAN	12000	0.3	100	80
58	148 Gerald	Cambrault	GCAMBRAU	011.44.134...	15/10/07	SA_MAN	11000	0.3	100	80
59	149 Eleni	Elotkey	EELOTKEY	011.44.134...	29/01/08	SA_MAN	10500	0.2	100	80
60	150 Peter	Tucker	PTUCKER	011.44.134...	30/01/05	SA_REP	10000	0.3	145	80
61	151 David	Bernstein	DBERNSTE	011.44.134...	24/03/05	SA_REP	9500	0.25	145	80
62	152 Peter	Hall	PHALL	011.44.134...	20/08/05	SA_REP	9000	0.25	145	80
63	153 Christopher	Giles	CGILES	011.44.134...	30/03/06	SA_REP	8000	0.2	145	80
64	154 Nanette	Cambrault	NCAMBRAU	011.44.134...	09/12/06	SA_REP	7500	0.2	145	80
65	155 Oliver	Tuvault	OTUVAULT	011.44.134...	23/11/07	SA_REP	7000	0.15	145	80
66	156 Janette	King	JKING	011.44.134...	30/01/04	SA_REP	10000	0.35	146	80
67	157 Patrick	Sully	PSULLY	011.44.134...	04/03/04	SA_REP	9500	0.35	146	80
68	158 Billie	McPhee	BMPHEE	011.44.134...	01/08/04	SA_REP	9000	0.35	146	80

JOBS

	JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
1	AD_PRES	President	20080	40000
2	AD_VP	Administration Vice President	15000	30000
3	AD_ASST	Administration Assistant	3000	6000
4	FI_MGR	Finance Manager	8200	16000
5	FI_ACCOUNT	Accountant	4200	9000
6	AC_MGR	Accounting Manager	8200	16000
7	AC_ACCOUNT	Public Accountant	4200	9000
8	SA_MAN	Sales Manager	10000	20080
9	SA_REP	Sales Representative	6000	12008
10	PU_MAN	Purchasing Manager	8000	15000
11	PU_CLERK	Purchasing Clerk	2500	5500
12	ST_MAN	Stock Manager	5500	8500
13	ST_CLERK	Stock Clerk	2008	5000
14	SH_CLERK	Shipping Clerk	2500	5500
15	IT_PROG	Programmer	4000	10000
16	MK_MAN	Marketing Manager	9000	15000
17	MK_REP	Marketing Representative	4000	9000
18	HR_REP	Human Resources Representative	4000	9000
19	PR_REP	Public Relations Representative	4500	10500

JOB_HISTORY

	EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
1	102	13/01/01	24/07/06	IT_PROG	60
2	101	21/09/97	27/10/01	AC_ACCOUNT	110
3	101	28/10/01	15/03/05	AC_MGR	110
4	201	17/02/04	19/12/07	MRK_REP	20
5	114	24/03/06	31/12/07	ST_CLERK	50
6	122	01/01/07	31/12/07	ST_CLERK	50
7	200	17/09/95	17/06/01	AD_ASST	90
8	176	24/03/06	31/12/06	SA_REP	80
9	176	01/01/07	31/12/07	SA_MAN	80
10	200	01/07/02	31/12/06	AC_ACCOUNT	90

LOCATIONS

	LOCATION_ID	STREET_ADDRESS	POSTAL_CODE	CITY	STATE_PROVINCE	COUNTRY_ID
1	1000	1297 Via Cola di Rie	00989	Roma	(null)	IT
2	1100	93091 Calle della Testa	10934	Venice	(null)	IT
3	1200	2017 Shinjuku-ku	1689	Tokyo	Tokyo Prefecture	JP
4	1300	9450 Kamiya-cho	6823	Hiroshima	(null)	JP
5	1400	2014 Jaberwocky Rd	26192	Southlake	Texas	US
6	1500	2011 Interiors Blvd	99236	South San Francisco	California	US
7	1600	2007 Zagora St	50090	South Brunswick	New Jersey	US
8	1700	2004 Charade Rd	98199	Seattle	Washington	US

9	1800 147 Spadina Ave	MSV 2L7	Toronto	Ontario	CA
10	1900 6092 Boxwood St	YSW 9T2	Whitehorse	Yukon	CA
11	2000 40-5-12 Laogianggen	190518	Beijing	(null)	CN
12	2100 1298 Vileparle (E)	490231	Bombay	Maharashtra	IN
13	2200 12-98 Victoria Street	2901	Sydney	New South Wales	AU
14	2300 198 Clementi North	540198	Singapore	(null)	SG
15	2400 8204 Arthur St	(null)	London	(null)	UK
16	2500 Magdalen Centre, The Oxford Science Park	CX9 9ZB	Oxford	Oxford	UK
17	2600 9702 Chester Road	09629850293	Stretford	Manchester	UK
18	2700 Schwanthalerstr. 7031	80925	Munich	Bavaria	DE
19	2800 Rua Frei Caneca 1360	01307-002	Sao Paulo	Sao Paulo	BR
20	2900 20 Rue des Corps-Saints	1730	Geneva	Geneve	CH
21	3000 Murtenstrasse 921	3095	Bern	BE	CH
22	3100 Pieter Breughelstraat 837	3029SK	Utrecht	Utrecht	NL
23	3200 Mariano Escobedo 9991	11932	Mexico City	Distrito Federal, MX	

REGIONS

	REGION_ID	REGION_NAME
1	1	Europe
2	2	Americas
3	3	Asia
4	4	Middle East and Africa

